MOHAMMAD MAHABOOB ASHRAF

Software & AI Engineer

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Computer Science undergraduate and AI Research Intern building evaluated AI systems, full-stack products, and reliable backend services. Open-source contributor to Agenta, Pydantic Logfire, and DSPy.

EDUCATION

Osmania University, Hyderabad — Expected Aug 2028

B.E. Computer Science Engineering | **CGPA: 8.9/10**

PROJECTS

Amana - Agentic Commerce Gateway | GitHub / Live — Aug 2026 - Present

Java, Spring Boot, Next.js, React, TypeScript, PostgreSQL/pgvector, Gemini, Razorpay

- Built a full-stack **agentic commerce system**: Gemini interprets multimodal intent and plans; deterministic Spring Boot services control authorization, payment execution, reconciliation, and refunds.
- Improved valid-product **Recall@5 from 55% to 76%** on 80 labelled cases; achieved **92% intent-field accuracy** across 52 utterances and validated **250/250 deterministic safety cases** through automated CI.

RepoPilot - Evidence-Grounded Coding Agent | GitHub — Aug 2026 - Sep 2026

Python, FastAPI, Tree-sitter, BM25, Chroma, LangGraph, React/TypeScript, Docker

- Built a local-first repair agent with hybrid repository retrieval, **hash-bound human approvals**, scoped patches, and Docker-isolated testing; passed **33/33 deterministic safety scenarios**.
- On the same six frozen defects, hosted Gemini repaired **6/6** versus local CPU Gemma's **1/6**; median workflow latency fell from **111s to 21s** with retrieval, prompts, and approval policy held fixed.

HookRelay - Reliable Webhook Delivery Platform | GitHub — Aug 2026

TypeScript, Node.js, Fastify, PostgreSQL, Redis/BullMQ, React, Docker

- Built durable webhook delivery with signed requests, bounded retries, dead-letter recovery, idempotent ingestion, and a React dashboard; **PostgreSQL owns delivery truth**, while Redis/BullMQ coordinates work.
- Verified idempotency under **25 parallel ingestion and replay requests**; measured **100% terminal success** across 3 x 500 local deliveries at **71-79 deliveries/second**.

OPEN SOURCE

Agenta | PR #7701 | **Merged** — Oct 2026

- Merged **bounded HTTPX streaming** for Slack-media downloads, enforcing attachment limits on decoded bytes before buffering each chunk and closing oversized responses early.
- Added regression coverage for missing or understated metadata, gzip responses, exact byte limits, and stream cleanup; **216 local Slack tests passed**.

Pydantic Logfire | PR #2650 | Open PR — Oct 2026

- Implemented separate root traces for FastAPI background-task batches, linked to the original request through **OpenTelemetry span links**, enabling independent tail-sampling decisions.
- Preserved callback ordering, exceptions, cancellation, and context restoration; documented scope and tested all four request/background sampling combinations. **35 focused tests passed**.

DSPy | PR #10563 | Open PR — Oct 2026

- Proposed exact prerelease-docs cleanup with **atomic stable publication**, immutable snapshots, and retry safety; **24 versioned-documentation tests passed**.

EXPERIENCE

Drugparadigm - AI Research Intern | Hyderabad — Oct 2025 - Present

- Developed antibody heavy/light-chain pairing models using protein-sequence embeddings over **1.3M+ heavy-chain sequences**; built preprocessing, leakage detection, and training-data validation pipelines.
- Outperformed ImmunoMatch on the same held-out benchmark: **Kappa 0.798 vs 0.750 ROC-AUC; Lambda 0.823 vs 0.783**. Benchmark analysis also exposed shortcuts in synthetic negative-pair construction.

iHub-Data, IIIT Hyderabad - AI Research Intern — Aug 2025 - Oct 2025

- Fine-tuned **YOLOv8m** on 1,350+ annotated UAV images, achieving **96.7% mAP@50, 94.7% precision, and 93.2% recall**; applied targeted augmentation and AdamW tuning.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript/JavaScript, SQL
Full Stack: React, Next.js, Vite, Node.js, Fastify, Spring Boot, FastAPI, REST APIs, PostgreSQL, Redis, BullMQ
AI / ML: LLMs, Agentic Workflows, Tool Calling, RAG, AI Evaluation, Embeddings, LangGraph, PyTorch, Gemini, Ollama
Engineering: Codex, Docker, GitHub Actions, Testing, Data Pipelines, API Integrations, Git
Fundamentals: Data Structures & Algorithms, OOP, DBMS